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| **Fiscal Year:** | FY 2016 | **Task Last Updated:** | FY 01/31/2016 |
| **PI Name:** | Alwood, Joshua Ph.D. | | |
| **Project Title:** | 2012 Presidential Early Career Award for Scientist and Engineers (PECASE) | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
| **PI Email:** | joshua.s.alwood@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 650-604-1490 |
| **Organization Name:** | NASA Ames Research Center | | |
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| **City:** | Moffett Field | **State:** | CA |
| **Zip Code:** | 94035 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Space Biology PECASE AWARD |
| **Start Date:** | 04/01/2015 | **End Date:** | 09/30/2018 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 3 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Taylor, Elizabeth | **Contact Phone:** | 650.604.1783 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | This Presidential Early Career Award for Scientist and Engineers (PECASE) award consist of two mini-projects. Project 1. Epigenetic Regulation During Simulated Spaceflight: MicroRNA Expression, Histone Deacetylase Status, and Methylation-Specific Sequencing<br><br>Description and Aim: This work complements the methylation work performed in the current Space Biology New Investigator Grant ("Bone Loss During Simulated Weightlessness: The Role of Osteoclasts," ends April 2015), by allowing ‘omics approaches to cover larger swaths of the genome (e.g., whole genome sequencing and RNA sequencing). Additionally, this project will extend the type of epigenetic mechanisms to be studied beyond DNA methylation. Specifically, we will analyze microRNA expression and status of histone acetylation. Hypotheses: Removal | | |

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| **Task Description:** | of mechanical loading or, oppositely, elevations of mechanical loading induce pro-bone resorbing or pro-bone forming changes in bone cell gene expression, respectively, via epigenetic mechanisms.<br><br>Project 2. Skeletal Sex Hormone Receptors and Simulated Spaceflight<br><br>Description and Aim: This work complements an upcoming space mission (PI : Joe Tash) and will define the skeletal response of male or female mice to simulated long-duration spaceflight in the context of sex hormones and their receptor levels in each compartment of the femur (cancellous and cortical). The assays used will include RNA-Seq, protein quantification in bone and blood, and immunostaining of histological sections of the femur for sex hormone and receptor levels. Hypotheses: Mechanical disuse reduces the levels of circulating sex hormones and their receptor levels in bone. These changes contribute to skeletal deterioration during long-duration spaceflight. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project addresses multiple Research Recommendations for Animal Experiments and Gaps in Knowledge stated in the Space Biology Science Plan and the National Academy of Sciences Decadal Survey, Recapturing Space: Life and Physical Sciences for a New Era of Exploration. The knowledge generated by this research queries the molecular responses to the high-turnover bone loss that astronauts experience with simultaneous weightlessness and exercising. This unique environment has similarities to Earth-bound sedentary or bed-ridden individuals. Additionally, this work investigates fundamental molecular mechanisms, specifically, non-coding RNA and DNA methylation responses to simulated spaceflight, that may improve our fundamental understanding of skeletal regulation and the organism's response to spaceflight. |
| **Task Progress:** | In the first year of the award period, progress has been made on each of the two projects listed in the Task Description. For Project 1, two experiments have been conducted. One experiment investigated the effects of musculoskeletal disuse, as a model of microgravity exposure, on non-coding microRNAs within the skeleton and the other experiment investigated the effects of compressive force (like those experienced during exercise) on the DNA methylation levels within bone.<br>For Project 2, one experiment was conducted to assess the temporal changes in gene expression levels of the sex-hormone receptors and osteoblastogenic transcription factors within bone following musculoskeletal disuse. Further, we are measuring the sex-hormone levels in the serum and plan to correlate all data. Finally, we have initiated pilot studies to develop an assay for fatigue properties of cancellous-rich bones in the axial skeleton for employment in future studies. |
| **Bibliography Type:** | Description: (Last Updated: 05/25/2026) |
| **Abstracts for Journals and Proceedings** | Scott RT, Nalavadi MO, Shirazi-Fard Y, Castillo AB, Alwood JS. "Effects of Zoledronate and Mechanical Loading During Simulated Weightlessness on Bone Structure and Mechanical Properties." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015.<br>31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015 |
| **Abstracts for Journals and Proceedings** | Thomas NJ, Stevick RJ, Tran LH, Nalavadi MO, Almeida EA, Globus RK, Alwood JS. "Does simulated spaceflight modify epigenetic status during bone remodeling?" Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015.<br>31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015 |
| **Abstracts for Journals and Proceedings** | Choi CY, Thomas NJ, Ronca-Finco AE, Christenson LK, Tash JS, Alwood JS. "The Effects of Simulated Weightlessness on Sex-Hormone Receptors in Bone During Bone Remodeling." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015.<br>31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015 |
| **Abstracts for Journals and Proceedings** | Pendleton MM, Alwood JS, O'Connell GD, Keaveny TM. "Design of a Fatigue Test for Ex-Vivo Mouse Vertebra." To be presented at 2016 Summer Biomechanics, Bioengineering and Biotransport Conference, National Harbor, MD, June 29-July 3, 2016.<br>SB3C2016. To be presented at 2016 Summer Biomechanics, Bioengineering and Biotransport Conference, National Harbor, MD, June 29-July 3, 2016. In press as of February 2016. , Feb-2016 |
| **Abstracts for Journals and Proceedings** | Scott RT, Nalavadi MO, Shirazi-Fard Y, Castillo AB, Alwood JS. "Effects of Zoledronate and Mechanical Loading During Simulated Weightlessness on Bone Structure and Mechanical Properties." Presented at the 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016.<br>2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016 |
| **Articles in Peer-reviewed Journals** | Alwood JS, Shahnazari M, Chicana B, Schreurs AS, Kumar A, Bartolini A, Shirazi-Fard Y, Globus RK. "Ionizing radiation stimulates expression of pro-osteoclastogenic genes in marrow and skeletal tissue." J Interferon Cytokine Res. 2015 Jun;35(6):480-7. Epub 2015 Mar 3. https://doi.org/10.1089/jir.2014.0152 ; PubMed PMID: 25734366; PubMed Central PMCID: PMC4490751 , Jun-2015 |
| **Awards** | Alwood JS. (Joshua S. Alwood) "Thora W. Halstead Young Investigator Award 2015. Awarded by American Society for Gravitational and Space Research, November 2015." Nov-2015 |
| **NASA Technical Documents** | Alwood JS, Mains RC. "Bone Physiology Introduction." In "Translational Cell and Animal Research in Space: 1965-2011." Ed. A.E. Ronca, K.A. Souza, R.C. Mains. NASA/SP-2015-625.<br>https://www.nasa.gov/sites/default/files/atoms/files/nasa-sp-2015-625.pdf , Sep-2015 |